Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Alex King	alxking@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

Help tool that helps students calculate stress, strain, and safety factor for a simple mechanical component.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

turn2engineering.com

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

calculate stress, strain, and safety factor for a simple mechanical component.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The ultimate goal of my program is to help users calculate normal stress and factor of safety for a mechanical component.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Applied force, Cross-sectional Area, Material Yield Strength, Target Factor of Safety

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It will use applied force and cross sectional area to calculate normal stress, then divide the materials yield strength to determine the factor of safety. It will also compare the result with the users target safety and show the results, and then a graph to show how the changes in force affect the components

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

Users will receive the calculated normal stress, the factor of safety, a comparison with their target safety, and a graph to help understand the mechanical components.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Component, Material, Force, Stress, Factor of safety, Target factor of safety, and calculation result

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Applied force on the component, Cross-sectional area of the component, Material yields strength, Target factor of safety. Details I will stay away from are the color and appearance.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The coding and making sure everything runs and works properly every time you use the program.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will keep asking questions about the code to ensure I avoid code that does nothing.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

Get a Good grade

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I am a mechanical engineering student creating a Factor of Safety Calculator for my introductory computer science final project. What are three common problems students face when calculating stress and factor of safety, and what are three simple features I could include in my web application to help solve these problems?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

Students struggle with remembering all the formulas, keeping constant units, and understanding how changing the applied force affects a component. Some fixes would be an automatic stress and factor of safety calculator, input validation, and a graph of how the safety changes with applied force. Solutions are online engineering calculators, spreadsheets, and professional engineering software.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I think a 3-d simulator is way too much for me as an introductory programming project so give me a simpler alternative. A simpler alternative would be to create a 2D graph that shows how changes in applied force affect the factor of safety.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

This has helped me make it more simple instead of a complex simulation. I will now keep focus on the calculating parts more than the simulation.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

LLM, Office hours, and Tutoring

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

Go through every line of code and if i cant find it ask a friend or the LLM to find the error.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

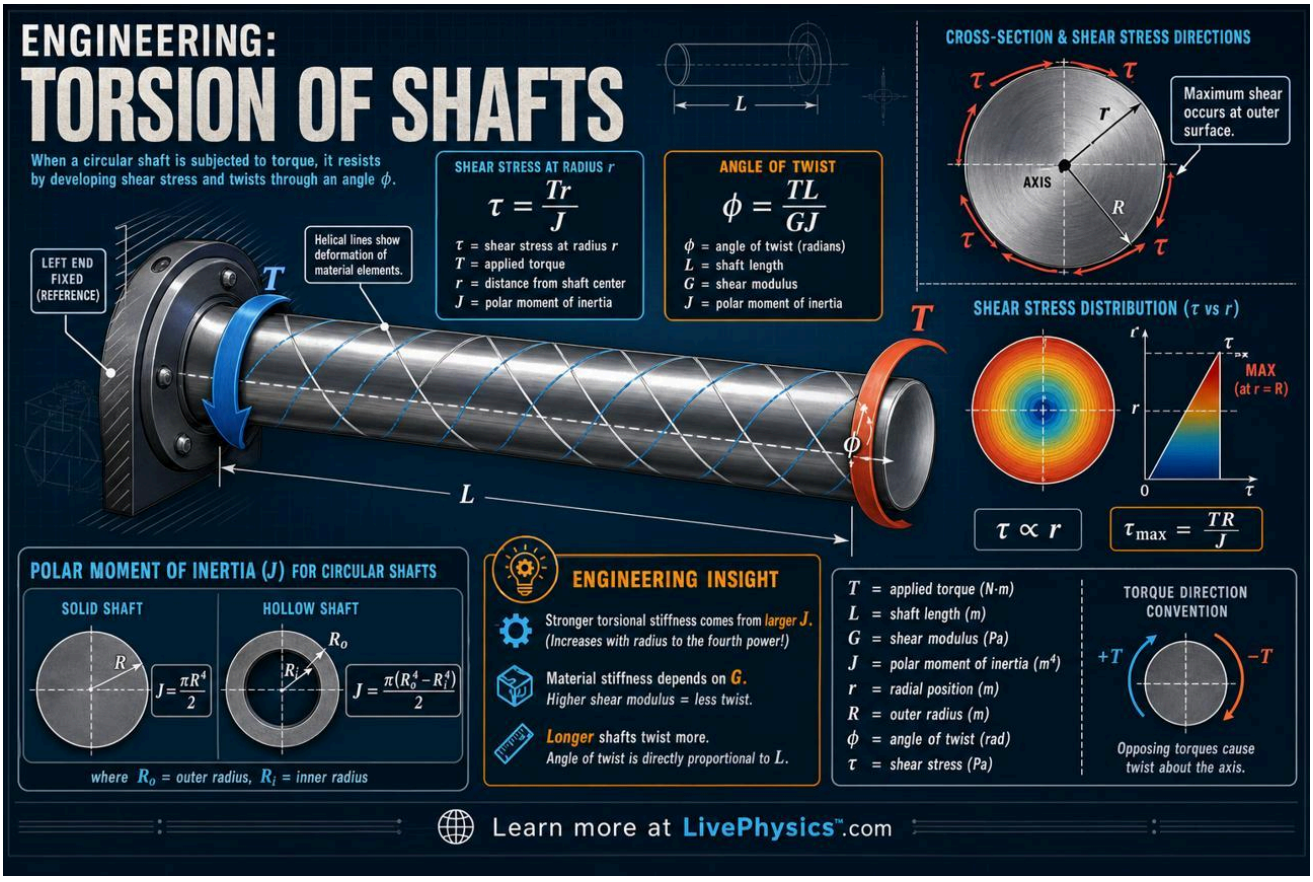
I am a mechanical Engineer what would a good project be for my final

As a mechanical engineer, I'd recommend building a project that combines real engineering calculations with an interactive web application. That way, your final project connects to your major, solves a practical problem, and demonstrates that you can program.

I looked at your CS1 Final Project worksheet. Since this is a three-week introductory programming project using Drafter, you want something with clear inputs, calculations, and outputs—not a full engineering simulation or a complicated system that requires a backend.

CS1 Final Project Alpha F26.pdf

My top 5 project ideas for a mechanical engineer

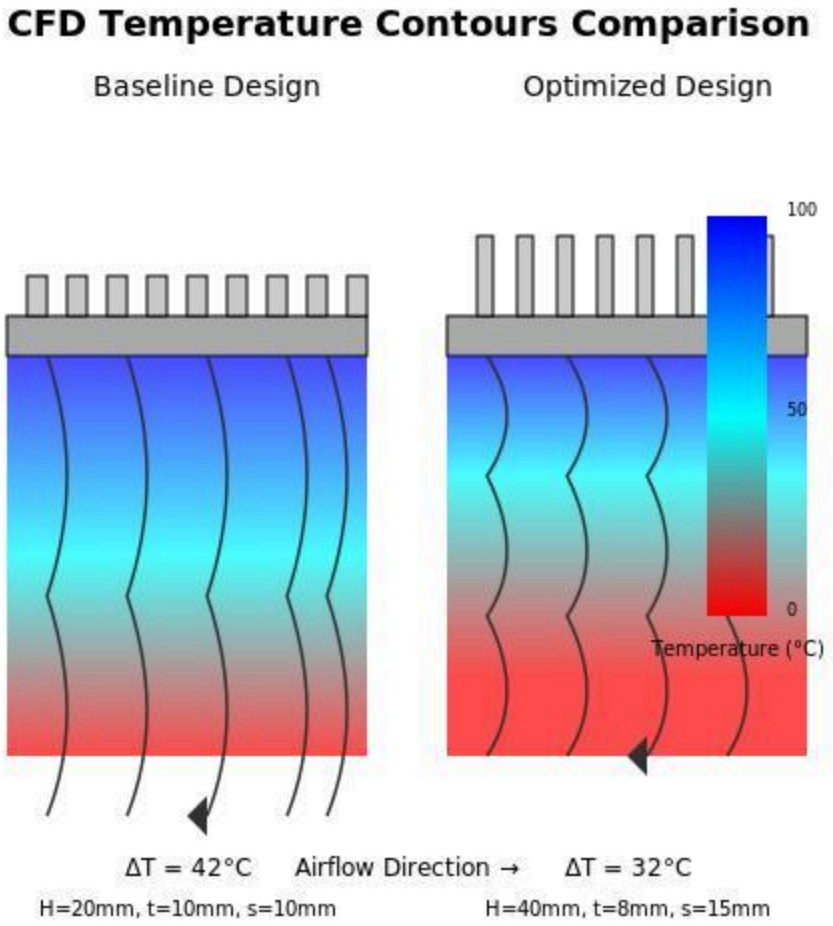


1. Engineering Design Calculator

My top pick

Build a tool that helps students calculate stress, strain, and safety factor for a simple mechanical component.

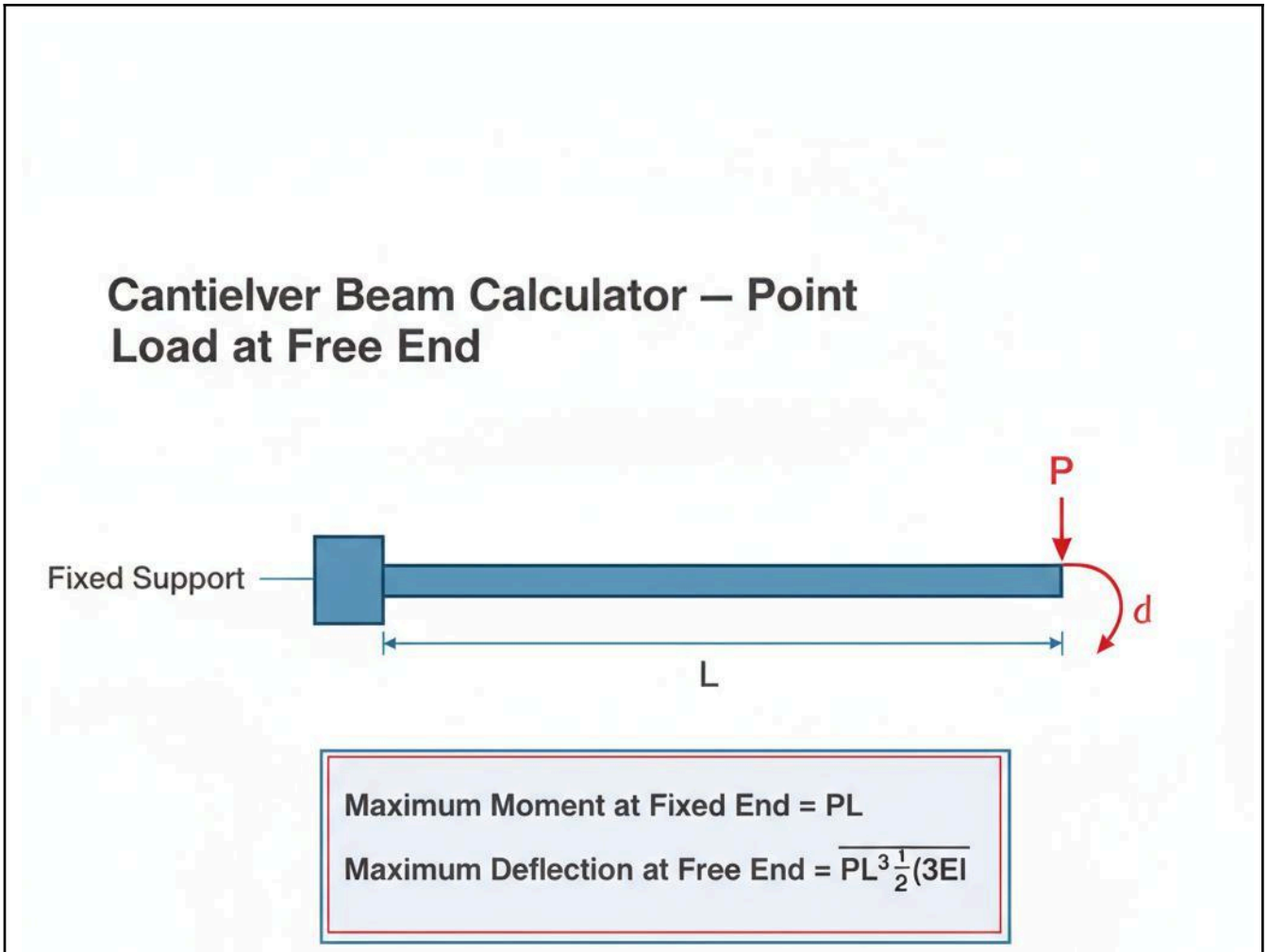
- Input: Force, cross-sectional area, material strength.
- Processing: Calculate stress and factor of safety.
- Output: Results, a safety assessment, and a graph.
- Why it works: It demonstrates core mechanical engineering concepts and uses basic Python math and conditional logic.



2. Heat Transfer Calculator

Create an app that estimates heat transfer through a wall or flat material.

- Input: Thermal conductivity, area, thickness, and temperature difference.
- Processing: Apply Fourier's law of heat conduction.
- Output: Heat-transfer rate and comparisons between materials.
- Why it works: It connects thermodynamics and heat transfer to an easy-to-use interface.



3. Beam Deflection Visualizer

Let users explore how a beam bends under different loads.

- Input: Beam length, load, elastic modulus, and cross-section dimensions.
- Processing: Calculate deflection using a standard beam formula.
- Output: Estimated deflection and a graph comparing different loads.

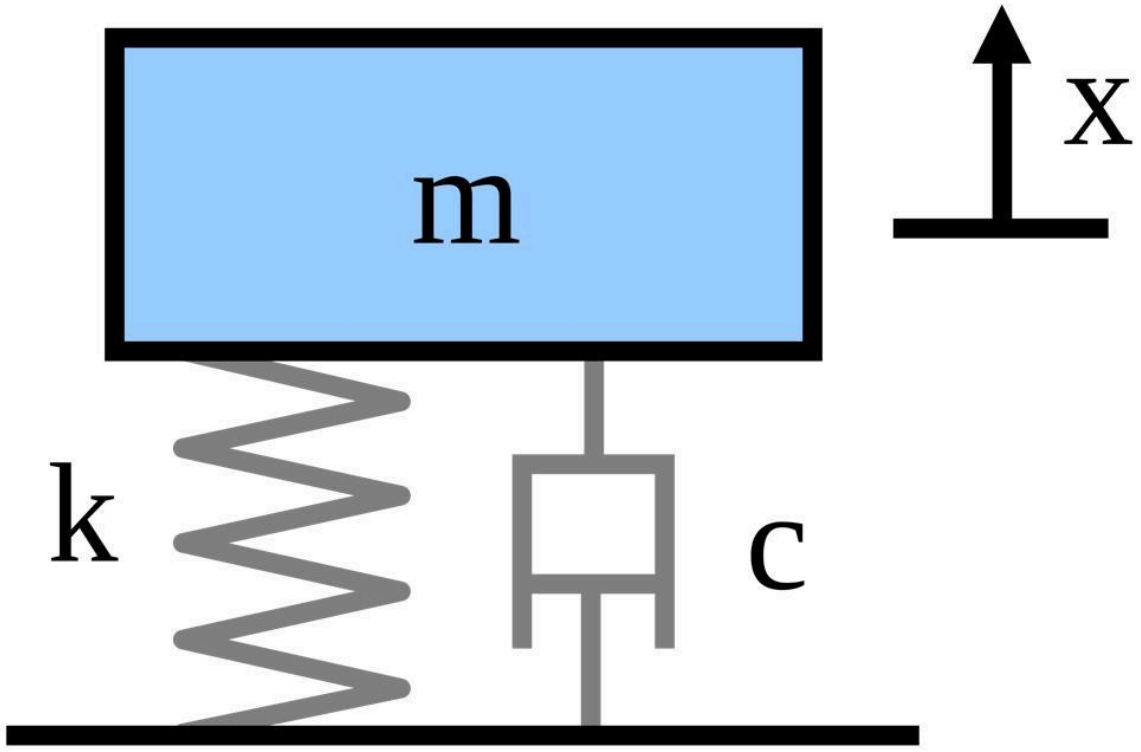
- Why it works: It makes mechanics of materials visual and interactive.



4. Gear Ratio and Speed Calculator

Build a tool for analyzing a simple two-gear system.

- Input: Driver and driven gear tooth counts, input speed.
- Processing: Calculate gear ratio and output rotational speed.
- Output: Output RPM, speed increase or reduction, and a comparison chart.
- Why it works: Straightforward to code, easy to test, and useful in machine design.



5. Vibration and Spring Calculator

Explore how a spring-mass system responds to changes in mass and stiffness.

- Input: Mass and spring stiffness.
- Processing: Calculate natural frequency.
- Output: Frequency in hertz and comparisons across different configurations.
- Why it works: It connects dynamics to interactive data visualization without needing a complex simulation.

Which project would I choose?

I would choose Project 1: Engineering Design Calculator, with a focused scope: a *Factor of Safety Calculator for Simple Mechanical Parts*.

It gives you a good balance of engineering depth, manageable Python programming, and an impressive demonstration.

Here is how it fits your assignment:

Worksheet section	Your project
Problem	Engineering students need a simple way to understand how applied loads affect a component's safety margin.
Prior solutions	Engineering textbooks, online stress calculators, and CAD/FEA software.
Your solution	An interactive calculator for estimating stress and factor of safety.
Input	Applied force, area, and material yield strength.
Processing	Calculate stress and factor of safety using formulas.
Output	Numerical results, a safety-category indicator, and a graph.
Data abstraction	A `Component` with force, area, and yield strength properties.
Technical challenge	Input validation, unit consistency, and displaying calculated results.

What the finished app could look like

Mechanical Design Assistant

Illustrative calculator preview

Applied force (N)

Cross-sectional area (mm²)

Material yield strength (MPa)

Normal stress

50.00 MPa

Factor of safety

5.00

Illustrative calculation for a uniform cross-section under direct axial loading. This is an educational estimate, not a design certification.

This is a demonstration of the kind of interaction you could implement in Drafter. The underlying calculations are simple:

$$\sigma = \frac{F}{A}$$

$$\sigma = \frac{F}{A}$$

$$\text{Factor of Safety} = \frac{\text{Yield Strength}}{\text{Applied Stress}}$$

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Because

$1\ \text{N/mm}^2 = 1\ \text{MPa}$

$1\ \text{N/mm}^2 = 1\ \text{MPa}$, these inputs give consistent stress units.

You could then add a matplotlib graph showing how the factor of safety changes as the

applied force increases.

Keep the scope manageable: start with direct axial loading, one material-strength input, and the two calculations. Add the graph and material presets only after the basic app works. A factor of safety alone does not establish that a real component is safe; actual design also depends on loading conditions, failure modes, fatigue, geometry, and applicable engineering standards.

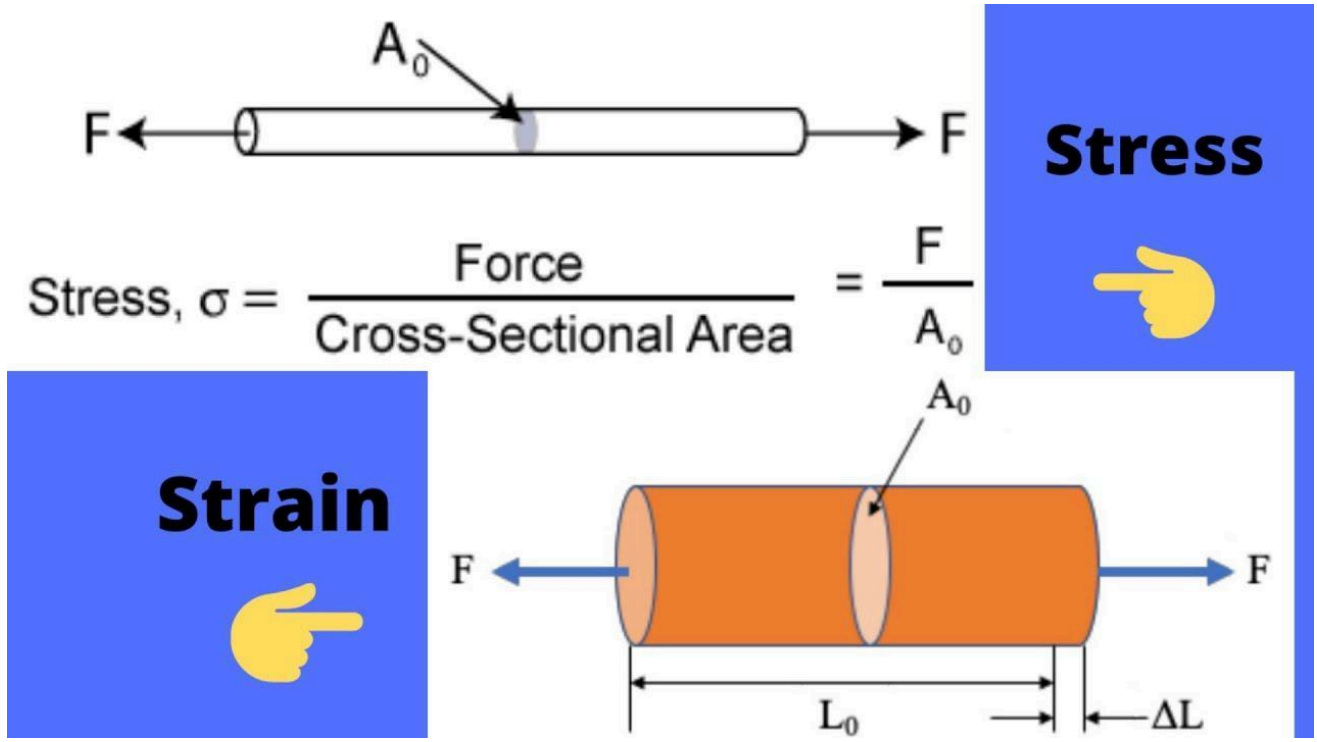
One question that would help me tailor the recommendation: would you rather build something related to your mechanical engineering coursework or something closer to a real-world engineering tool?

I can narrow the project to fit your interests and the three-week CS1 deadline.

I will choose the first one

Great choice! Let's develop Project 1: Engineering Design Calculator — Factor of Safety Calculator for Mechanical Components into a plan that fits your CS1 final project and your mechanical engineering background.

Your project concept



Your web application will allow a user to enter an applied force, the component's cross-sectional area, and the material's yield strength. The application will calculate the stress and factor of safety, then display the results in an easy-to-understand format.

Features to build

1. Engineering calculations

Calculate normal stress and factor of safety using basic Python arithmetic.

2. Safety assessment

Display the calculated factor of safety and let the user compare it with a user-selected target. Clearly label this as a simplified educational assessment.

3. Interactive graph

Use matplotlib to show how stress or factor of safety changes as the applied force

increases.

4. Input validation

Prevent invalid inputs, such as zero cross-sectional area or negative material strength.

help me brainstorm

For your mechanical engineering Factor of Safety Calculator, your brainstorming section should explore the problem you're solving, what features would be useful, and what you can realistically build in three weeks.

Part 5: Research & Brainstorming Conversation

1. The Initial Prompt/Question

You can use this prompt for your worksheet:

I am a mechanical engineering student creating a Factor of Safety Calculator for my introductory computer science final project. What are three common problems students face when calculating stress and factor of safety, and what are three simple features I could include in my web application to help solve these problems?

I think a 3-d simulator is way to much for me as an introductory programming project so give me a simpler alternative

A simpler alternative would be to create a 2D graph that shows how changes in applied force affect the factor of safety. The user would enter the force, cross-sectional area, and material yield strength, and the program would calculate the factor of safety and display the results in a graph. This would be much easier to develop using basic Python and matplotlib while still demonstrating an important mechanical engineering concept.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.